Appendix: An Integrated Rigid-Flexible Body Dynamic Approach to Computationally Efficient Musculoskeletal Modeling and Muscle Recruitment Simulation of the Lumbosacral Spine and Torso

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This appendix describes the models provided in the GitHub repository, instructions to simulate the models and other interesting methods and results that could not be included in the original manuscript.

I. Model Repository

The repository can be accessed at: https://github.com/siril-teja/ROMmodel-io.git. It contains the main project file named *ROMmodel-io.prj* which can be opened in MATLAB. Note that Simulink 2023a or a newer version is required to run this model. To run any optimization, Simulink Design Optimization or Control System Toolbox is needed.

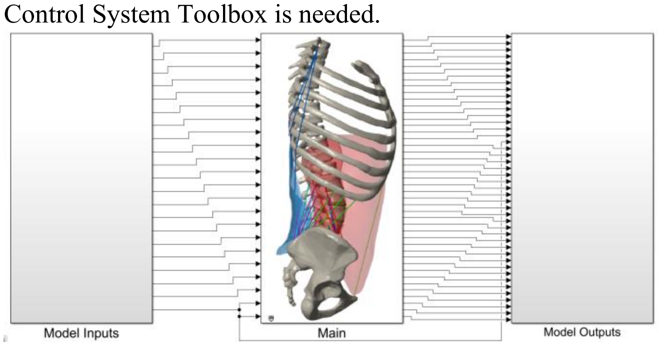

Fig. 1. Model schematic

Four separate Simulink models are included in the project to facilitate simulation of different cases discussed in the original manuscript (C0_C3, C1, C2, C4_C5).

- Case 0 = Use ROMmodel_C0_C3.slx with muscle force inputs set to zero.
- Case 1 = Use ROMmodel_C1.slx
- Case 2 = Use ROMmodel_C2.slx
- Case 3 = Use ROMmodel_C0_C3.slx with desired muscle force inputs.
- Case 4 = Use ROMmodel_C4_C5.slx with muscle force inputs set to zero.
- Case 5 = Use ROMmodel_C4_C5.slx with desired muscle force inputs.

All the Simulink models contain three main blocks. The model input and output blocks are editable and are used to pass arguments and fetch responses to and from the main model. The user can add more inputs and derive new output measurements by editing these blocks accordingly. The main model is not editable in the current version.

A. Model Inputs

Model inputs facilitate simulating the model in different loading scenarios. They can either be a constant value or a time varying signal and. All the tunable inputs of the model are listed below.

TABLE I
TUNABLE MODEL INPUTS

Parameter name	Description (units: N, Nm)
FollowerLoad	Follower compression load on the lumbar spine
Right_Lat	Right Latissimus Dorsi muscle force
Left_Lat	Left Latissimus Dorsi muscle force
Right_Psoas	Right Psoas Major muscle force
Left_Psoas	Left Psoas Major muscle force
Right_MF	Right Multifidus muscle force
Left_MF	Left Multifidus muscle force
Right_LongThor	Right Longissimus Thoracis muscle force
Left_LongThor	Left Longissimus Thoracis muscle force
Right_EO	Right External Oblique muscle force
Left_EO	Left External Oblique muscle force
Right_IO	Right Internal Oblique muscle force
Left_IO	Left Internal Oblique muscle force
Right_RA	Right Rectus Abdominus muscle force
Left_RA	Left Rectus Abdominus muscle force
Moment_FE	Flexion-extension moment at L1 centroid
Moment_LB	Lateral bending moment at L1 centroid
Moment_AR	Axial rotation moment at L1 centroid
T1 Vertical	Compressive force at T1 centroid
T1 Horizontal	Flexion force at T1 centroid
L1 Horizontal	Flexion force at L1 centroid

TABLE II
FIXED MODEL INPUTS

Parameter name	Description
IAP_activation_AbdFlex	IAP activation %. It is set to 100% by default. Relevant in cases 1, 4 and 5.
E_TLFflex	Elastic modulus of TLF. It is set to 18.8 MPa by default. Relevant in cases 2, 4 and 5.
E_IVDn	Elastic modulus of L1 to L5 IVDs. They are set to 2.1, 2.2, 1.1, 1.7, 3.2 MPa by default as per the optimization results for zero follower load case discussed in section III.A of the original manuscript. Relevant in all cases.
other	Graphics related – not relevant.

The pelvis of the model is fixed in all degrees of freedom relative to the ground. Since the main model is protected, some input parameters that were discussed in the original manuscript needed to be fixed as constants. These parameters were directly referenced inside the Simulink blocks and could not be tuned from outside the main model. They are listed in Table II.

B. Model Outputs

An extensive list of model measurements was made available by default. Some of them are:

- L1-S1 moment and range of motion
- Trunk displacement and range of motion
- Segmental moments and range of motions
- IVD compressive and shear forces
- Force exerted by abdominal cavity on lumbar spine segments
- Cost functions
- Stability criterion: deflection from gravity plumbline

In addition to these, the user can create new outputs as desired using the signal lines provided.

All the simulation cases analyzed in the study are shown in this video.

C. Optimization

In addition to performing any simulations, the user has an ability to perform optimization using the tunable parameters as inputs and model outputs as targets. Simulink Design Optimization or Control Systems toolbox is needed for this. Once installed, the user can utilize the "Check Custom Bounds" blocks to model their constraints. Then, the inbuilt Response Optimizer app can be used to define optimization inputs and optimizer settings.

D. Model Benchmarking

To compare the computational demand of the model in all cases, a benchmarking script (`benchmarking_all.m`) was included in the repository. This script loads the model, sets the model parameters if any, depending on the case, and simulates the model. It outputs the model compile and run times for all the cases. The models were simulated 5 times and the average simulation times were reported in section III.F of the original manuscript.

II. Additional Methods

A. Muscle Recruitment Optimization Routine

To optimize the muscle forces for the following set of constraints, i.e. imposing stability criterion on the spine against an external perturbation, and the maximum muscle stress is less than 460 kPa, with the objective of minimizing the total muscle effort, a global optimization algorithm called surrogate optimization (`surrogateopt`) was used [1]. Six initial guesses (0N-250N) with 100 iterations per guess were evaluated and the trade off plot was plotted. The best solution in each of the 6 sets was chosen and used to plot the model moment-rotation curves in cases 3 and 5 (figure 10 of the original manuscript). The following shows the flowchart of the steps involved.

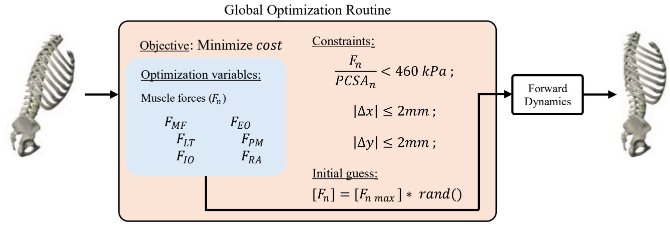

Fig. 2. Optimization routine used to derive muscle recruitment solutions for a given set of external loads to bring back the spine into stable region.

B. Abdominal Cavity Elasticity

The IAP activation in the model was controlled by varying the abdominal cavity elastic modulus to tune the resistance offered by the cavity under flexion. Cholewicki et al suggested that the increase in trunk stiffness due to 40% and 80% IAP in humans is 21% and 42% respectively [2]. Using these data points, the increase in trunk stiffness at 100% IAP was extrapolated to be 52.5% as shown in figure 3a. Then this data was translated into the model, by including the abdominal cavity and adjusting its elastic modulus until they met the prescribed ranges of motions. The elastic moduli at these points were recorded and are correlated with the activation %, as shown in figure 3b.

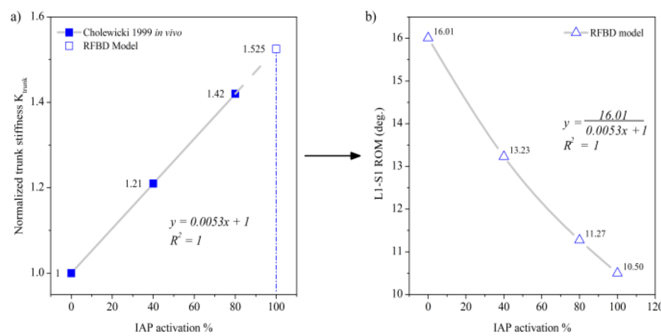

Fig. 3. a) Normalized trunk stiffness vs IAP activation as prescribed by Cholewicki et al. [2], and model stiffness extrapolated at 100% activation. b) Model range of motion (ROM) vs IAP activation % achieved by tuning the abdominal cavity elastic modulus.

III. Additional Results

A. IDP Validation

The model showed a linear increase in intradiscal pressure at L4-L5 with the increase in follower compressive load, matching well with the literature *ex vivo* [3] and *in silico* [4] models. Here the intra discal pressure was calculated by dividing the normal compressive force on the disc with its area of cross-section.

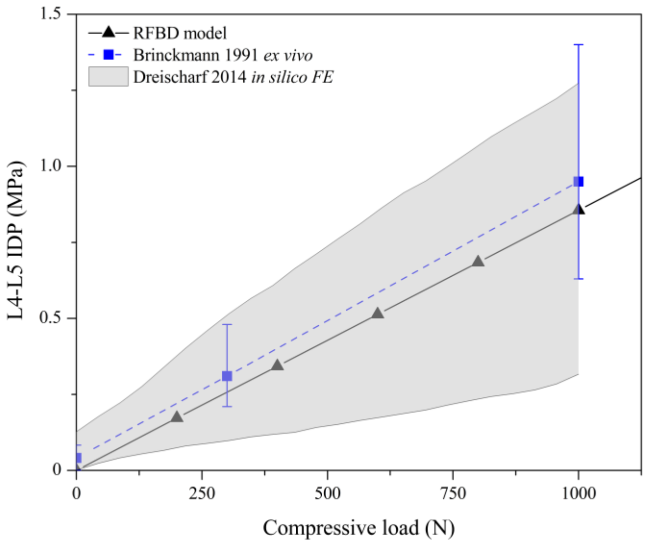

Fig. 4. Variation of L4-L5 intradiscal pressure with increase in compressive follower load in the RFBD model compared to ex vivo [3] and in silico models [4] in literature.

B. IVD Forces

The compressive and shear forces in the IVDs decreased exponentially with the increase in IAP activation. Notably, the compressive forces in all the discs reached near zero at 200% IAP activation and 7.5Nm flexion load at L1. Further increase in IAP resulted in normal tensile forces in the discs indicating that the abdominal cavity possibly acting as a lever against the external moment at L1, transferring the load on to the cavity through the ribcage-diaphragm interface.

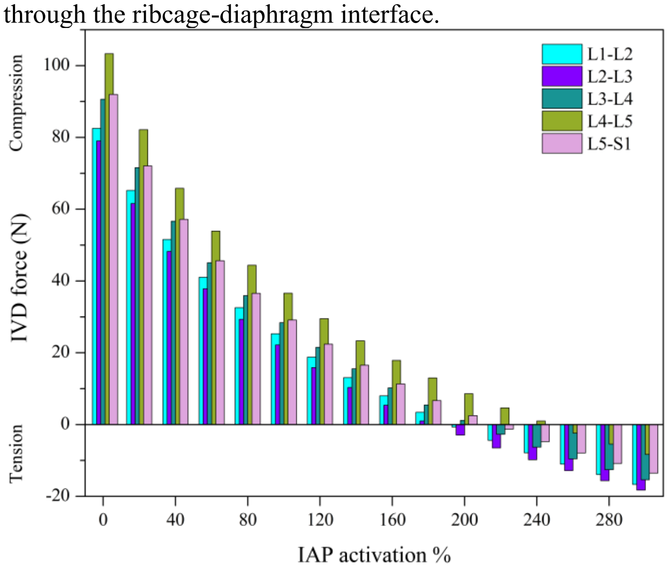

Fig. 5. Variation of IVD compressive forces with increase in IAP activation at 7.5Nm flexion moment.

The first three levels of the spine recorded an anterior shear, while the last two recorded shear in posterior direction, indicating an inflexion point about L3. The L5-S1 level recorded the highest shear forces out of all levels and continued to decrease with the increase in the IAP activation %. Level-by-level compressive and shear force breakdowns are provided in figures 5 and 6.

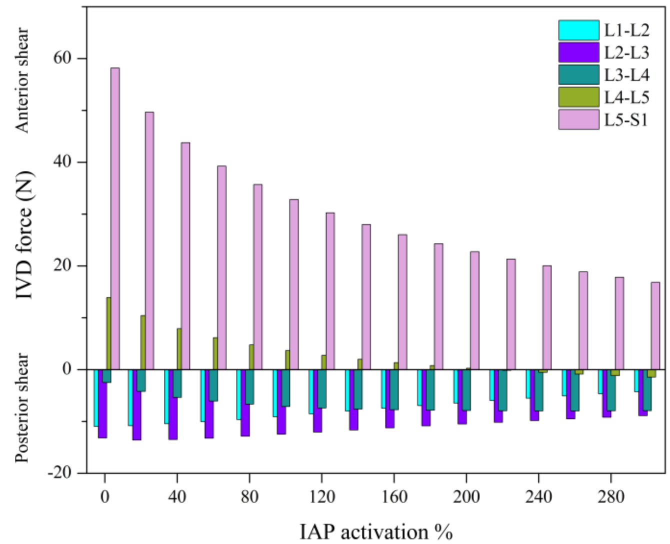

Fig. 6. Variation of IVD shear forces with increase in IAP activation at 7.5Nm flexion moment.

C. Comparison to Previous RBD Version [5]

The RFBD model presented in this study is a continued effort from our previous RBD spine model [5]. Both these models shared some similarities such as the modeling software MATLAB Simscape Multibody (The MathWorks Inc., MA, USA), the CAD geometry, and some loading conditions discussed in the original manuscript. The RBD model had 3 DOF spherical IVD joints, whereas the current RFBD model has the 6 DOF flexible body IVDs. The IAP in the RBD model was modeled using normal force vectors and spring-damper elements, while in the RFBD model, it was modeled using a flexible abdominal cavity between the diaphragm and the pelvic floor. As the spring constant of the IAP in the RBD model increased, the lumbar spine ROM decreased, and similarly with the IAP activation in the RFBD model. Both these approaches show similar results of increased trunk stiffness in flexion and the ROMs are compared in figure 5.

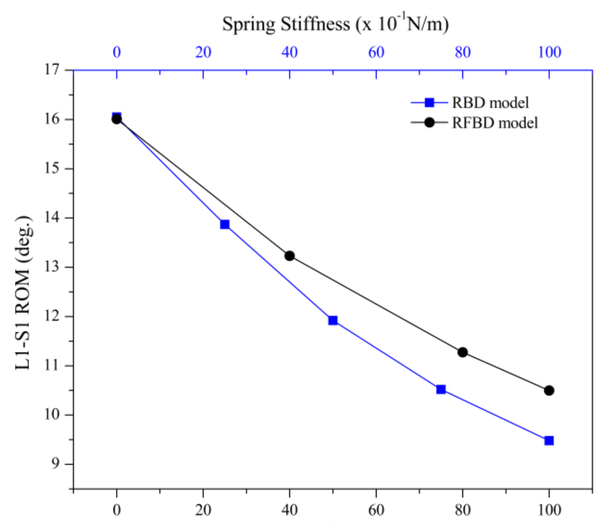

Fig. 7. Comparison of the IAP implementation of the RFBD model to its previous rigid-body version [5].

D. Model Calibration and Validation in other directions

The model was also calibrated and validated in extension, lateral bending and axial rotation directions, following the same E_{IVD} optimization procedure described in the section II B of the manuscript. Follower load was set to zero, and the target moment-rotation curves were derived from the polynomial functions given by Zhang et al. in [6]. Table III lists the optimized values for E_{IVD} in the three principal directions.

TABLE III
OPTIMIZED E_{IVD} VALUES (MPA)

Level	Flexion	Extension	Lateral Bending	Axial Rotation
L1-L2	2.1	3.2	1.7	21
L2-L3	2.2	3.5	1.7	23
L3-L4	1.1	1.9	0.7	14.5
L4-L5	1.7	4.5	1.6	18.5
L5-S1	3.2	6	4.4	36

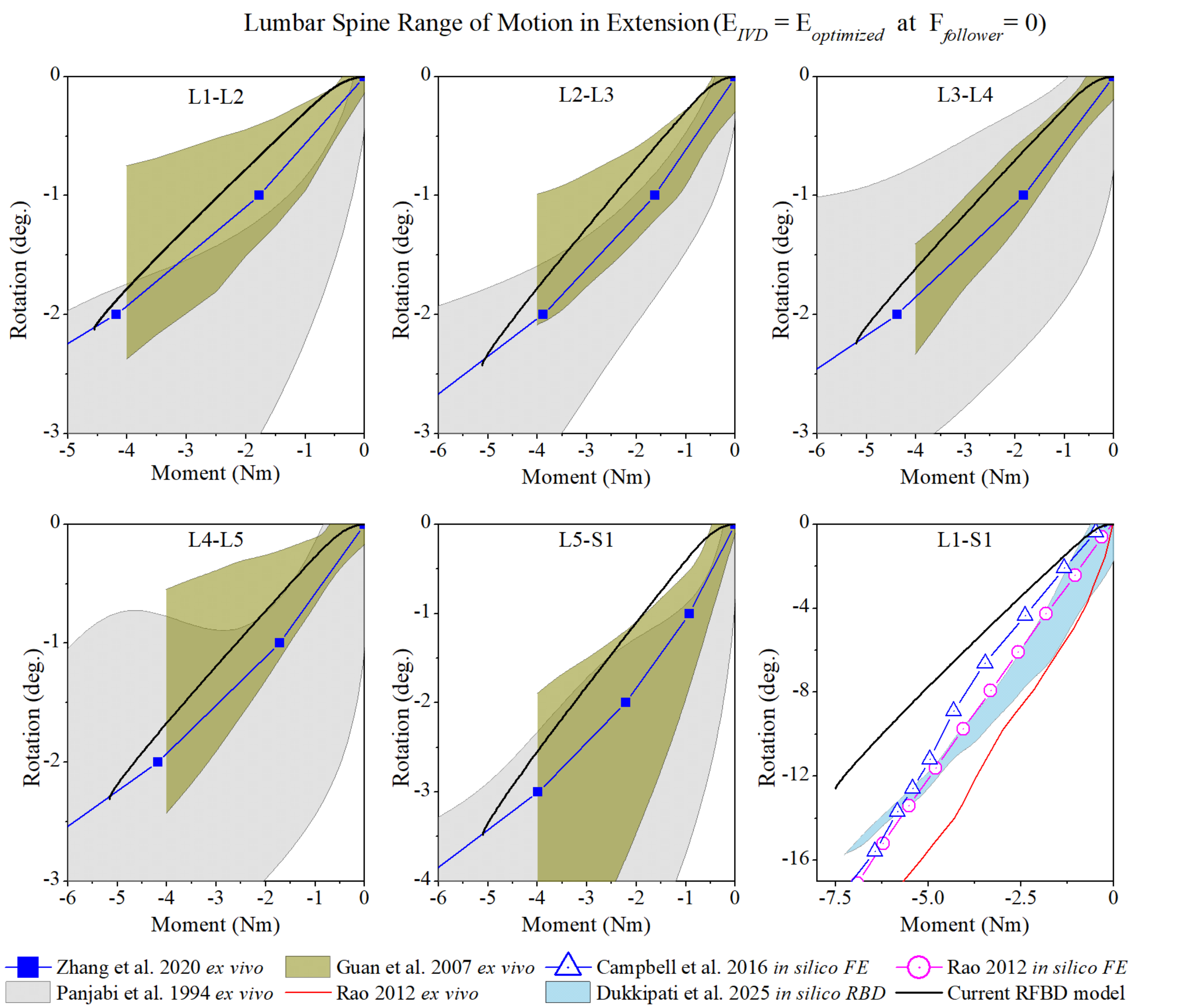

Fig. 8. Momen-Rotation curves of the lumbar spine in extension with optimized IVD elastic moduli at zero follower load case compared to literature *ex vivo* (Zhang et al. 2020 [6], Panjabi et al. 1994 [7], Guan et al. 2007 [8], Rao 2012 [9]), *in silico* finite element (Rao 2012 [9], Campbell et al. 2016 [10]), *in silico* rigid body (Dukkipati et al. 2025 [5]) models.

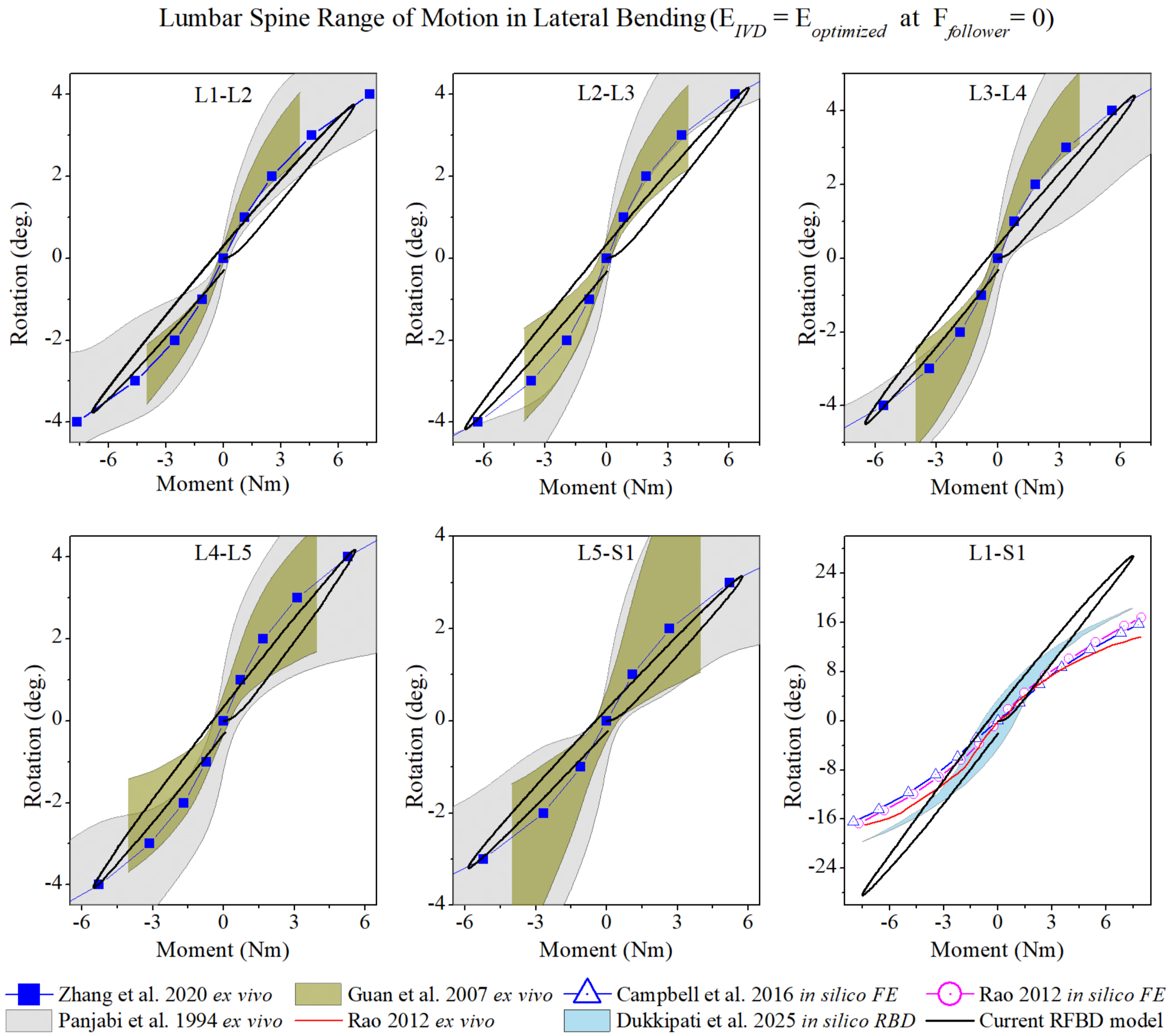

Fig. 9. Momen-Rotation curves of the lumbar spine in lateral bending with optimized IVD elastic moduli at zero follower load case compared to literature *ex vivo* (Zhang et al. 2020 [6], Panjabi et al. 1994 [7], Guan et al. 2007 [8], Rao 2012 [9]), *in silico* finite element (Rao 2012 [9], Campbell et al. 2016 [10]), *in silico* rigid body (Dukkipati et al. 2025 [5]) models.

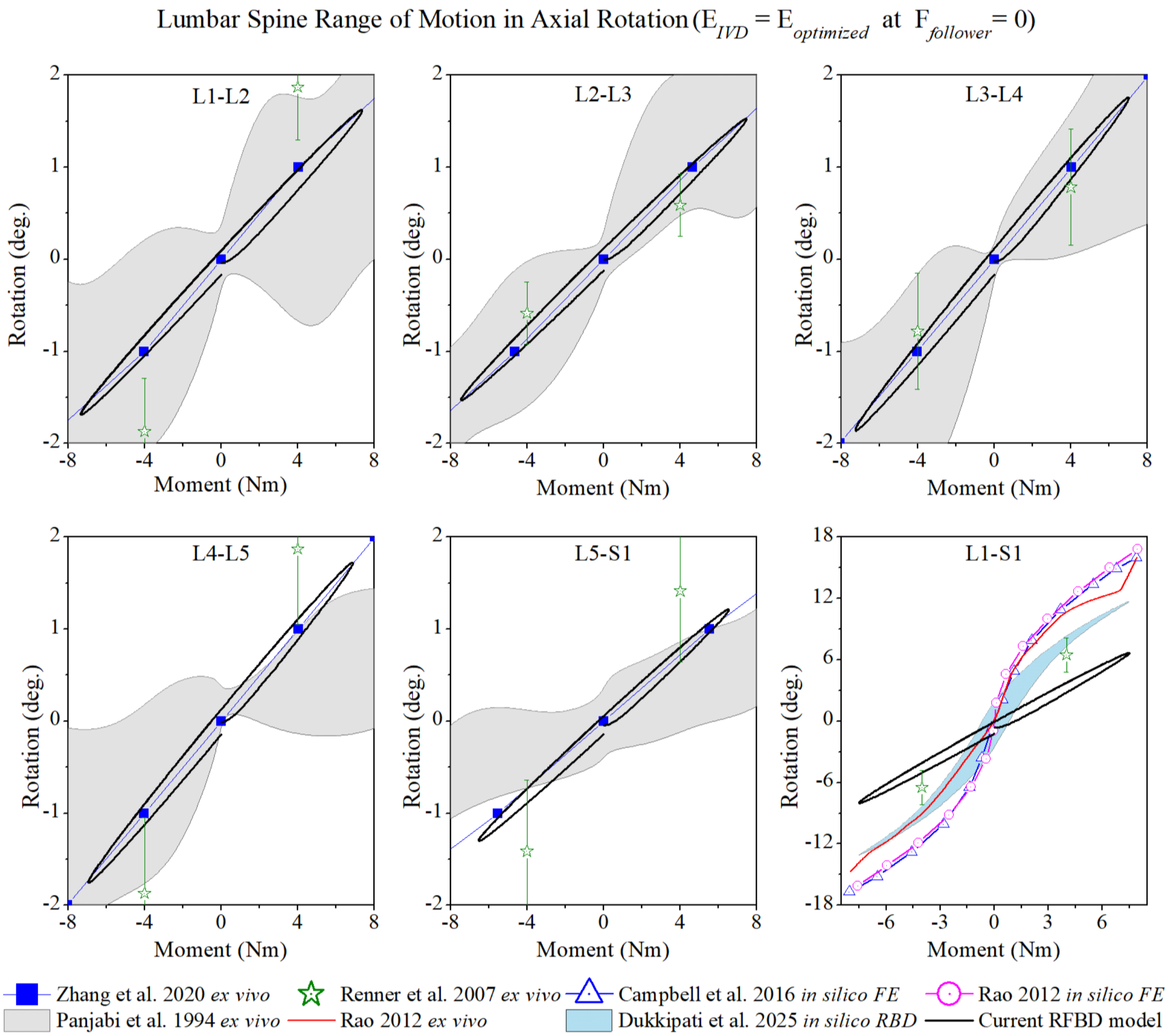

Fig. 10. Momen-Rotation curves of the lumbar spine in axial rotation with optimized IVD elastic moduli at zero follower load case compared to literature *ex vivo* (Zhang et al. 2020 [6], Panjabi et al. 1994 [7], Renner et al. 2007 [11], Rao 2012 [9]), *in silico* finite element (Rao 2012 [9], Campbell et al. 2016 [10]), *in silico* rigid body (Dukkipati et al. 2025 [5]) models.

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